

1. Set up the partial fraction decomposition of the following rational functions. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^3 - 2x^2 + 3x - 4}{x(x+1)^2(x^2+9)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{Dx+E}{x^2+9} + \frac{Fx+G}{(x^2+9)^2}.$$

2. Evaluate the improper integral $\int_0^{\infty} \frac{1}{(x+2)^3} dx$ or show that it is divergent.

(Note: Answers without appropriate limit notations will result in loss of points.)

$$u = x+2, \quad du = dx$$

$$\int \frac{1}{(x+2)^3} dx = \int \frac{1}{u^3} du = -\frac{1}{2} \cdot \frac{1}{u^2} + C = -\frac{1}{2} \cdot \frac{1}{(x+2)^2} + C$$

$$\int_0^{\infty} \frac{1}{(x+2)^3} dx = \lim_{t \rightarrow \infty} \int_0^t \frac{1}{(x+2)^3} dx = \lim_{t \rightarrow \infty} \left(-\frac{1}{2} \cdot \frac{1}{(x+2)^2} \Big|_0^t \right)$$

$$= \lim_{t \rightarrow \infty} \left(-\frac{1}{2} \cdot \frac{1}{(t+2)^2} - \left(-\frac{1}{2} \cdot \frac{1}{(0+2)^2} \right) \right)$$

$$= \lim_{t \rightarrow \infty} \left(-\frac{1}{2} \cdot \frac{1}{(t+2)^2} + \frac{1}{8} \right)$$

$$= \frac{1}{8}$$

$$\Rightarrow \int_0^{\infty} \frac{1}{(x+2)^3} dx = \frac{1}{8}, \text{ Convergent.}$$